

LINEAR ALGEBRA (MIS 203)

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MID TERM EXAM (FALL 2023)

CLASS # 7500.

Question-1

let $A\bar{x} = \bar{b}$ be a linear system whose augmented matrix $(A|B)$ has reduced row echelon form,

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

(a) Find all solution to the system.

It follows from the reduced row echelon form that the free variables are

$$x_2 = \alpha, \quad x_4 = \beta \quad \text{and} \quad x_5 = \gamma$$

$$\text{Then } x_1 = -2 - 2\alpha - 3\beta - \gamma$$

$$x_3 = 5 - 2\beta - 4\gamma$$

and hence the solution consists of all vectors of the form

$$\bar{x} = (-2 - 2\alpha - 3\beta - \gamma, \alpha, 5 - 2\beta - 4\gamma, \beta, \gamma)^T$$

(b) If $a_1 = [1, 1, 3, 4]^T$ and $a_3 = [2, -1, 1, 3]^T$ determine $\bar{b}$.

If we set the free variables equal to 0, then

$x_0 = (-2, 0, 5, 0, 0)^T$ is a solution to $A\bar{x} = \bar{b}$

and hence

$$\bar{b} = Ax_0 = -2a_1 + 5a_3 = (8, -7, -1, 7)^T$$

$$b = \begin{bmatrix} 1 & 2 & 2 & 3 & 1 \\ 1 & 0 & -1 & 2 & 4 \\ 3 & 0 & 1 & 0 & 0 \\ 4 & 0 & 3 & 0 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 5 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ -7 \\ -1 \\ 7 \end{bmatrix}$$

Question - 2

(2)

Let A be an $n \times n$ matrix and let

$$B = A + A^T \quad \text{and} \quad C = A - A^T$$

A is symmetric if its transpose is equal to itself, and skew symmetric if its transpose is equal to its negative. (a) Prove that B is symmetric and C is skew symmetric.

$$\because B = A + A^T$$

Taking transpose both sides

$$B^T = (A + A^T)^T = A^T + (A^T)^T = A^T + A = B$$

Hence B is symmetric.

$$\because C = A - A^T$$

$$C^T = (A - A^T)^T = A^T - (A^T)^T$$

$$= A^T - A$$

$$= -(A - A^T)$$

$$C^T = -C$$

Hence C is skew symmetric.

(b) Prove that every square matrix of order n can be written as a sum of a symmetric matrix and a skew symmetric matrix.

$$\text{For symmetric matrix: } A = A^T \Rightarrow A - A^T = 0$$

$$\text{For skew symmetric matrix: } A = -A^T \Rightarrow A + A^T = 0$$

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

(c) Validate Part (b) by giving an example of a square matrix of order 4×4 .

$$\text{Let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \text{ and } A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & a_{41} \\ a_{12} & a_{22} & a_{32} & a_{42} \\ a_{13} & a_{23} & a_{33} & a_{43} \\ a_{14} & a_{24} & a_{34} & a_{44} \end{bmatrix}$$

A matrix can be written as a sum of a symmetric and a skew symmetric matrix

i.e. $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = \text{Symmetric} + \text{Skew Symmetric}$

where A is symmetric i.e. $a_{21} = a_{12}$, $a_{31} = a_{13}$, $a_{41} = a_{14}$,
 $a_{23} = a_{32}$, $a_{34} = a_{43}$, $a_{24} = a_{42}$.

where A is skew symmetric.

$a_{21} = -a_{12}$, $a_{13} = -a_{31}$, $a_{41} = -a_{14}$, $a_{23} = -a_{32}$, $a_{34} = -a_{43}$,
 and $a_{24} = -a_{42}$.

$$= \frac{1}{2} \left\{ \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{12} & a_{22} & a_{23} & a_{24} \\ a_{13} & a_{23} & a_{33} & a_{34} \\ a_{14} & a_{24} & a_{34} & a_{44} \end{bmatrix} \right\} +$$

$$\frac{1}{2} \left\{ \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} - \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ +a_{12} & a_{22} & a_{23} & a_{24} \\ +a_{13} & +a_{23} & a_{33} & a_{34} \\ +a_{14} & +a_{24} & +a_{34} & a_{44} \end{bmatrix} \right\}$$

Question-3.

If A and B are $n \times n$ matrices, then prove that $\det(AB) = \det(A) \det(B)$. Use elementary matrices to show this, furthermore, do not imply any results on elementary matrices, Prove intermediate results on elementary matrices as and when required.

If B is singular

It follows that AB is also singular.

$$\therefore \det(AB) = 0 = \det(A) \det(B)$$

If B is nonsingular, then B can be written as a product of elementary matrices.

$$B = E_k E_{k-1} \dots E_2 E_1 I$$

$$\text{Thus } AB = A E_k E_{k-1} \dots E_2 E_1$$

$$\begin{aligned} \Rightarrow \det(AB) &= \det(A E_k E_{k-1} \dots E_2 E_1) \\ &= \det(A) [\det(E_k) \det(E_{k-1}) \dots \det(E_2) \det(E_1)] \\ &= \det(A) \det(E_k E_{k-1} \dots E_2 E_1) \end{aligned}$$

$$\det(AB) = \det(A) \det(B)$$

Proved

(5)

Question - 4

Let $A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$ where all four blocks are $n \times n$ matrices.

If A_{11} , and A_{22} are nonsingular, show that A must also be nonsingular, and that A^{-1} must be of the form

$$\left[\begin{array}{c|c} A_{11}^{-1} & C \\ \hline 0 & A_{22}^{-1} \end{array} \right] \quad \text{Determine } C.$$

$$\begin{aligned} A A^{-1} &= \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} A_{11}^{-1} & C \\ 0 & A_{22}^{-1} \end{bmatrix} \\ &= \begin{bmatrix} A_{11} A_{11}^{-1} + 0 & A_{11} C + A_{12} A_{22}^{-1} \\ 0 + A_{11}^{-1} 0 & 0 + A_{22} A_{22}^{-1} \end{bmatrix} \\ &= \begin{bmatrix} I & A_{11} C + A_{12} A_{22}^{-1} \\ 0 & I \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \end{aligned}$$

$$A_{11} C + A_{12} A_{22}^{-1} = 0$$

$$A_{11} C = -A_{12} A_{22}^{-1}$$

$$C = -A_{11}^{-1} A_{12} A_{22}^{-1}$$

and Therefore

$$A A^{-1} = I = A^{-1} A$$

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}$$

(6)

Question - 5

Perform the following block multiplication:

$$\left[\begin{array}{cc|cc} 3/5 & -4/5 & 0 & 0 \\ 4/5 & 3/5 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \end{array} \right] \left[\begin{array}{cc|c} 3/5 + 4/5 & 0 & 0 \\ -4/5 & 3/5 & 0 \\ \hline 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

$$\text{let } A = \left[\begin{array}{cc|cc} 3/5 & -4/5 & 0 & 0 \\ 4/5 & 3/5 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \end{array} \right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$\& B = \left[\begin{array}{cc|c} 3/5 & 4/5 & 0 \\ -4/5 & 3/5 & 0 \\ \hline 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right] = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}$$

$$= \left[\begin{array}{cc|c} 9/25 + 16/25 & 12/25 - 12/25 & 0 \\ -12/25 + 12/25 & 16/25 + 9/25 & 0 \\ \hline 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{cc|c} 25/25 & 0 & 0 \\ 0 & 25/25 & 0 \\ \hline 0 & 0 & 1 \end{array} \right]$$

$$AB = \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right]$$